

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
DLA 14

Lung Function & Respiratory Gases 3

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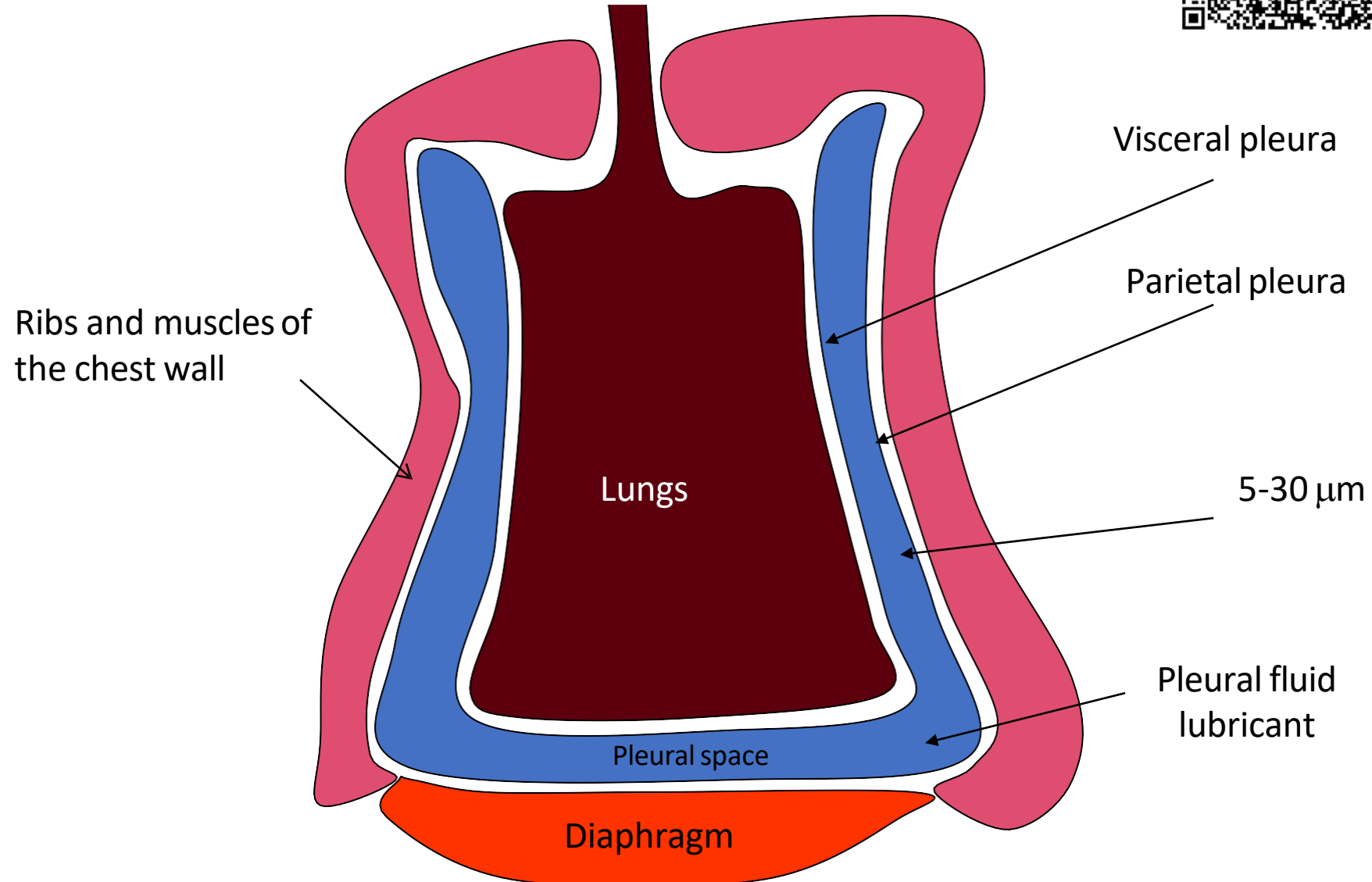
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Learning Objectives

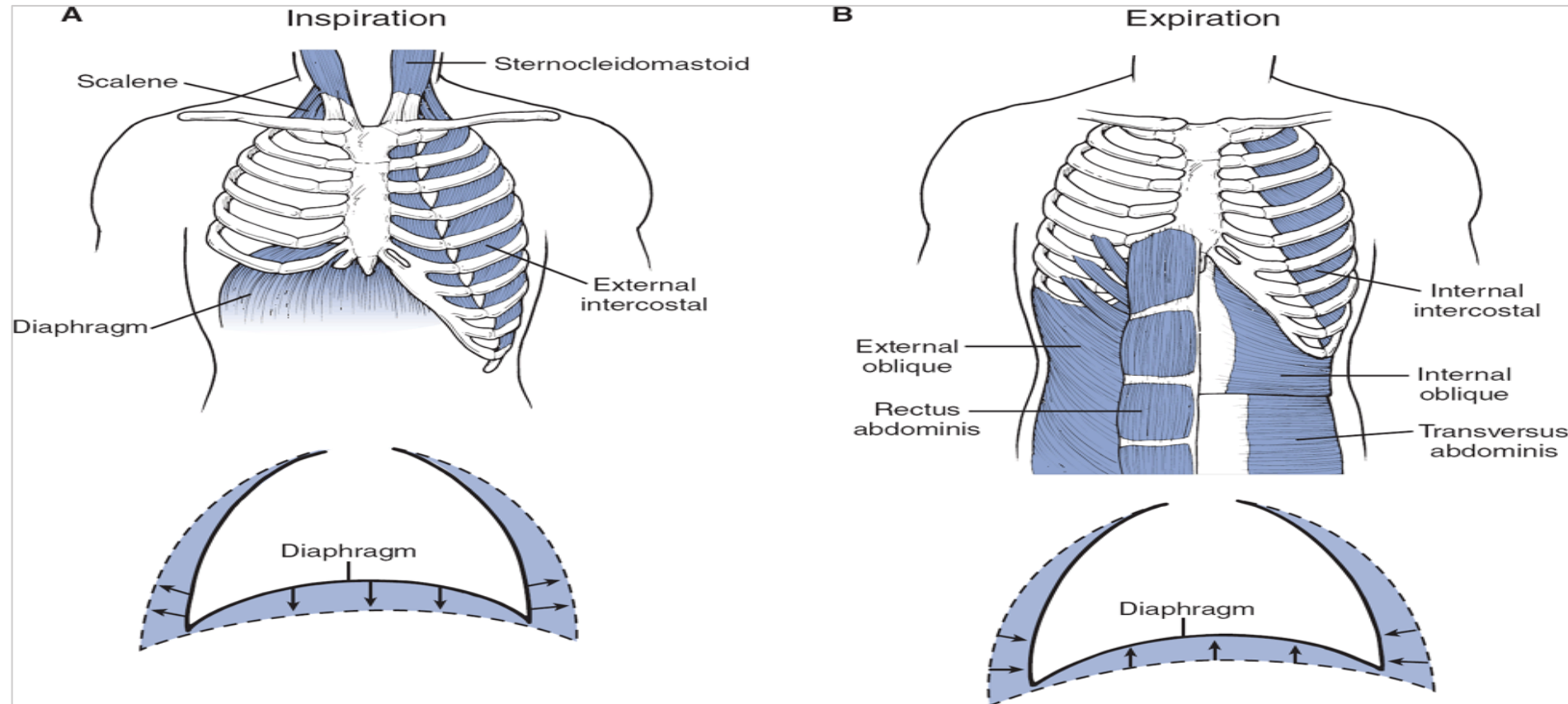
By the end of this DLA, the student should be able to

SOM.MK.II.BPM1.3.CPR.2.PHYS.0191	Describe the structural characteristics of airways.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0192	Describe the transport of gas through the conducting airways to and from the alveoli.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0193	Explain the key roles of mucociliary transport and describe related pathologies and their clinical presentation.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0194	Describe the clinical significance of mucociliary transport.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0195	Describe the functions of alveolar Type 1 & type II cells.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0196	List components of chest wall and lungs and relate the functions of respiratory muscles to movement of air in and out of the alveoli.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0197	Describe how Boyle's and Charles' law apply to movement of gas in and out of lungs.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0198	List the metabolic functions of the lungs and explain their key physiological roles.

Respiratory System – Chest Wall and Air Movement



Muscles of Respiration



Rhoades and Bell, Medical Physiology, p329

Illustration of muscles involved in the different phases of respiration.

Muscles of Respiration



Muscles of Inspiration	Muscles of Expiration
Diaphragm	Internal Intercostal
External Intercostal	Innermost Intercostal
Scalene	Rectus Abdominis
Sternocleidomastoid	Transverse Oblique
Pectoralis muscles	Internal Oblique
Serratus anterior	
Latissimus dorsi	

Table 1. List of muscles involved in Inspiration and Expiration

Pressure and Gas Movement

- Respiration has two phases – **Inspiration** and **Expiration**
- What causes the movement of air during these phases?
- **Air moves because of pressure gradients**

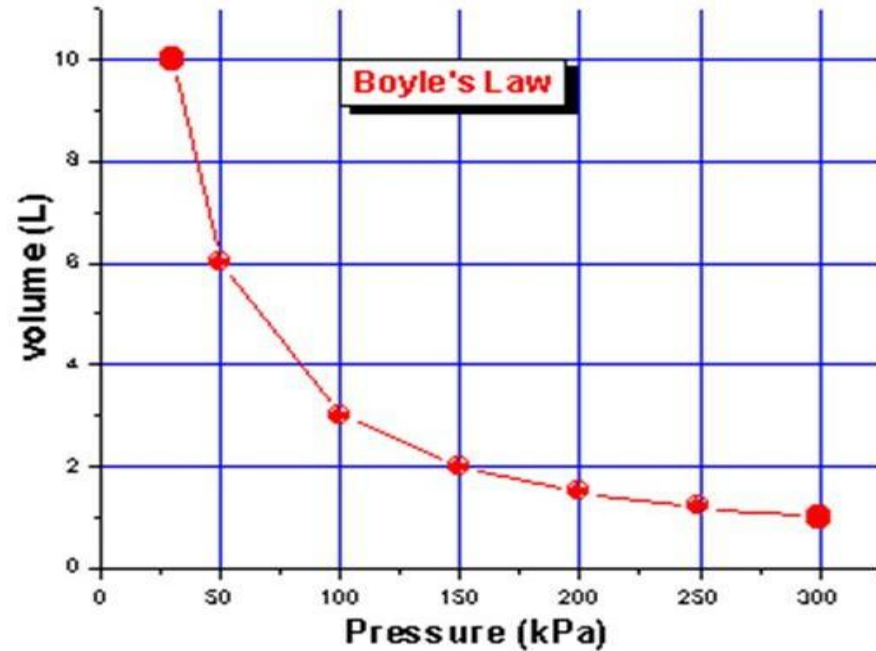


Boyle's Law

Charles Law

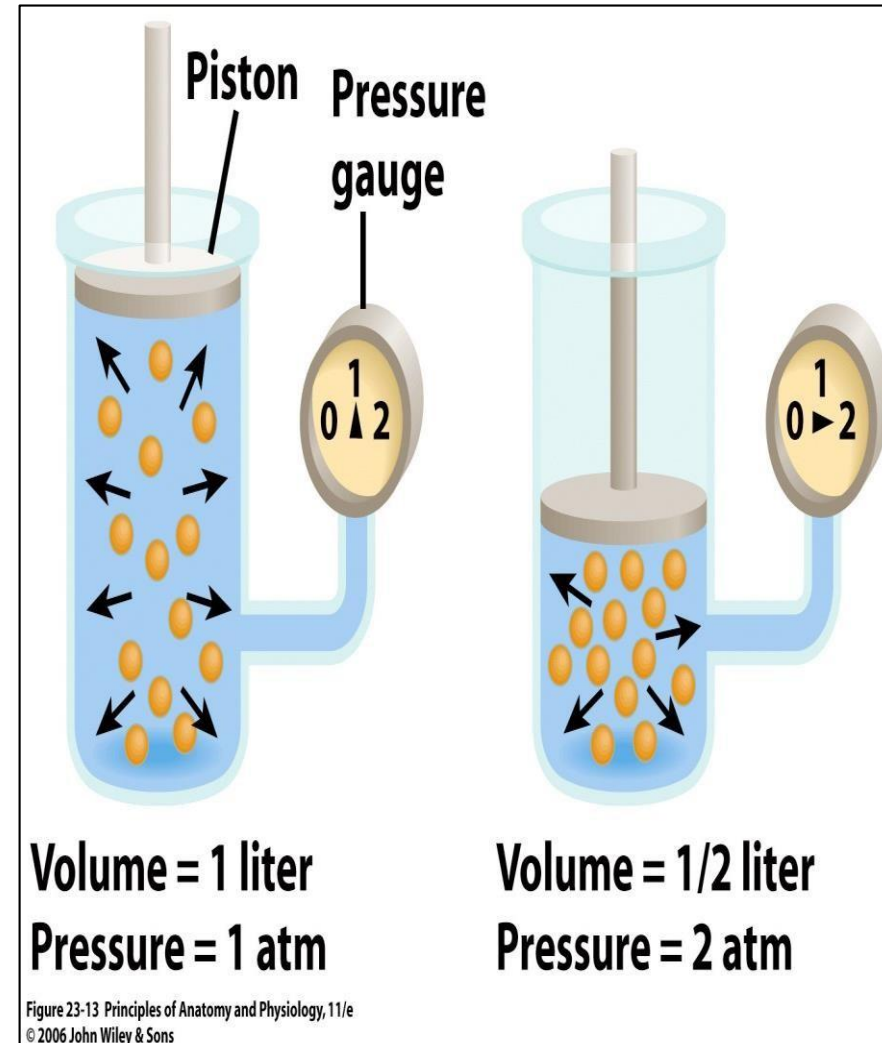
Pressure and Gas Movement

<http://www.volvoab.com/image/boyles-law/boyle-s-delight-and>



Graph showing the relationship between volume and pressure at fixed temperature.

Boyle's Law: The volume of a given mass of a fixed amount of gas is inversely proportional to its pressure provided the temperature remains constant



Pressure and Gas Movement

Charles' Law gives the relationship between volume and temperature in an ideal gas where the pressure is constant.

At constant pressure, volume (V) is directly proportional to the temperature (T):

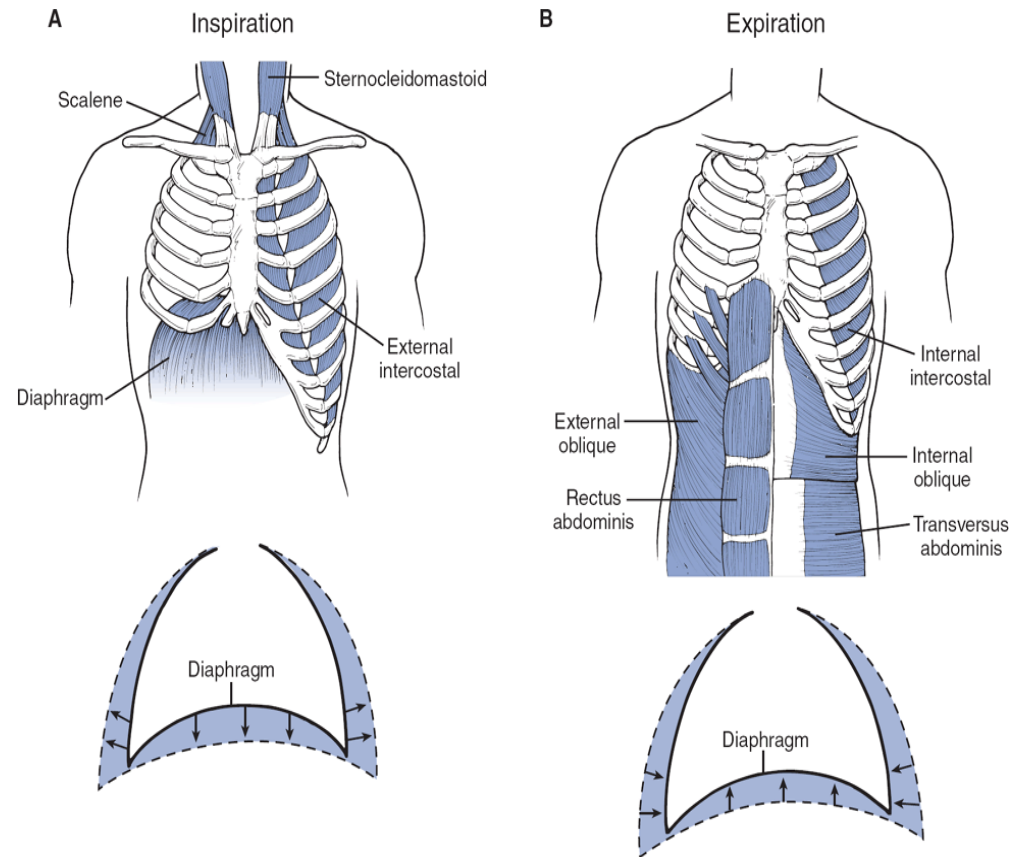
1) $V \sim T$

2) $V_1/T_1 = V_2/T_2$

At constant temperature, an increase in thoracic volume leads to decrease in pleural pressure

Muscles of Respiration

Airtight chest wall allows for creation of pressure changes that facilitates the movement of air on inspiration and expiration.



Rhoades and Bell, Medical Physiology, p329

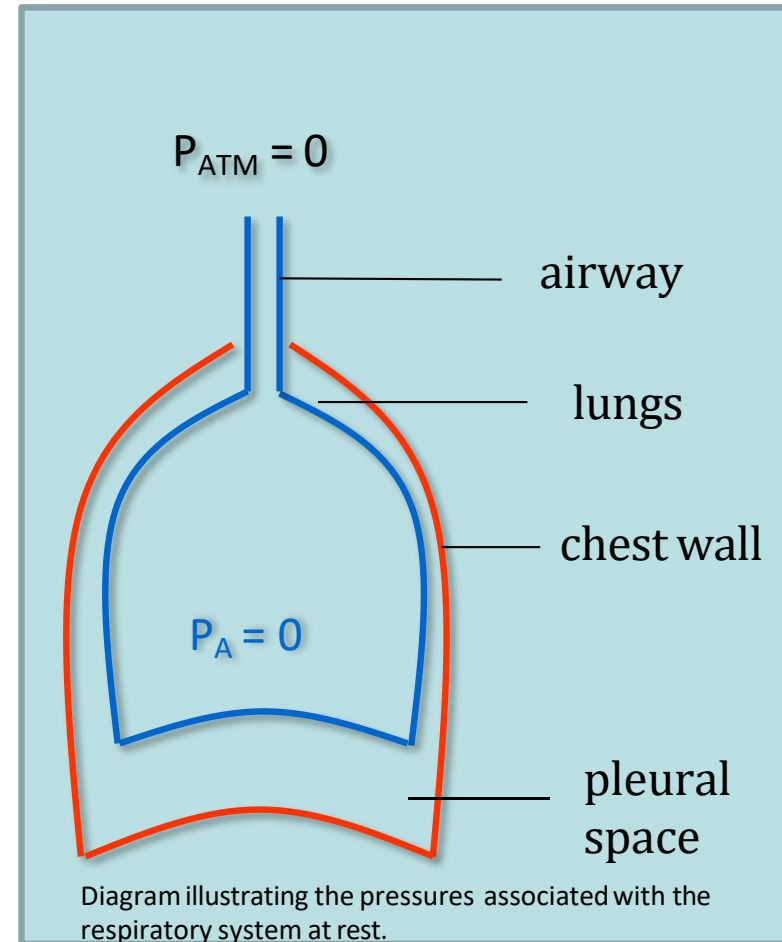
Pressures and Airflow

- **Atmospheric Pressure (P_{ATM})**
 - Pressure exerted by the earth's atmosphere
 - 760 mm Hg at sea level
- **Alveolar pressure (P_A)**
 - The pressure within the alveoli

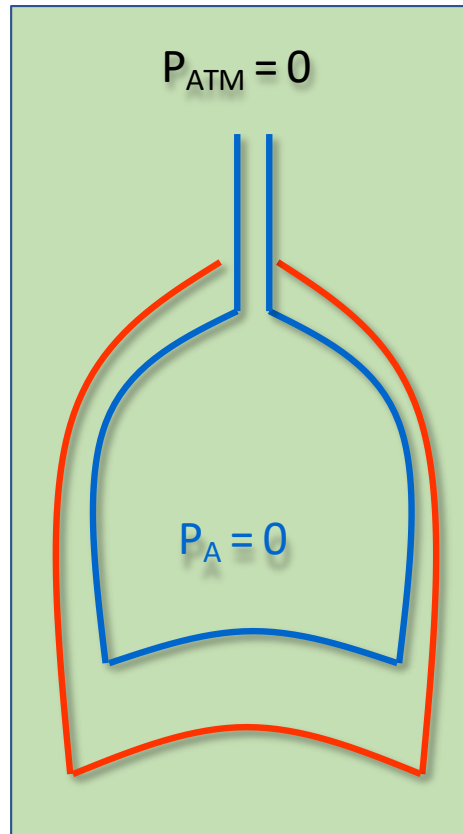
Pressures and Airflow

Air flows because of pressure gradients

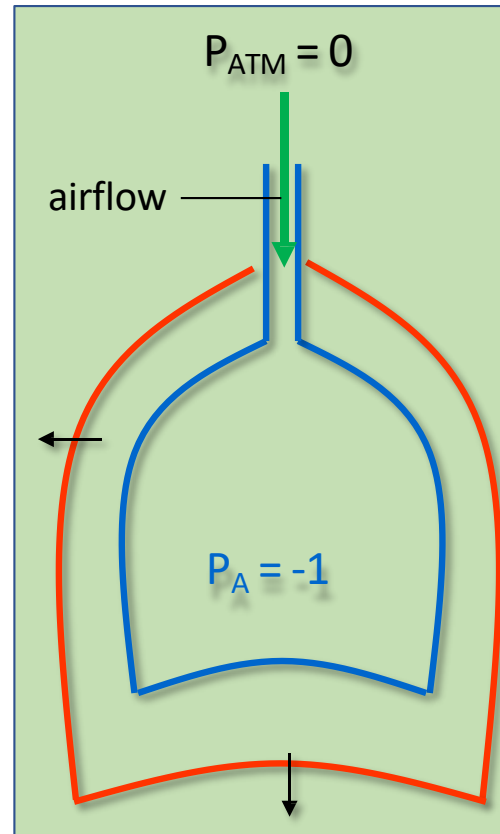
- Air flows into the lung when the alveolar pressure (P_A) falls below atmospheric pressure (P_{ATM})
- Air moves out of the lungs when P_A becomes greater than P_{ATM}
- What influences the changes in P_A ?



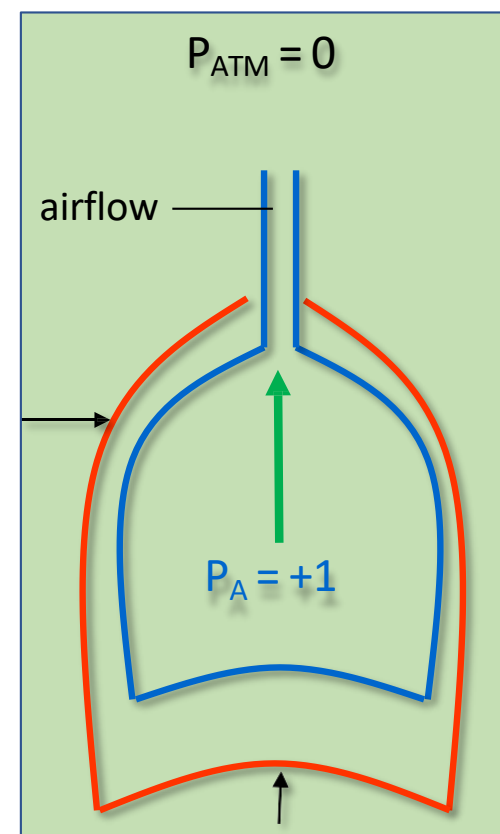
Pressure and Gas Movement



At Rest



Airflow on
Inspiration



Airflow on
Expiration

Summary

- During quiet respiration , inspiration is accomplished principally by contraction of the diaphragm and expiration is a passive process.
- The accessory muscles of respiration are utilized only with forced respiration.
- Air flows into and out of the respiratory tree because of the pressure gradient created by the changes in thoracic volume.

Review the information in this DLA by completing the table below

Phase of Respiration	Muscles involved	Changes in alveolar volume	Changes in alveolar pressure